DeepSeek – R1 – Zero

Reinforcement Learning →

↳ Model learn by trial & error

We don't generate example that the model can learn form like the supervised learning.

We just take the models output & do some kind of evaluation.

① Prompt template

↳ how example questions are passed to model

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A conversation between User and Assistant. The user asks a
question, and the Assistant solves it.The assistant first
thinks about the reasoning process in the mind and then
provides the user with the answer. The reasoning process
and answer are enclosed within <think> </think> and
<answer> </answer> tags, respectively, i.e., <think>
reasoning process here </think>
<answer> answer here </answer>. User: {prompt}. Assistant:
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② Rule-based reward

↳ The intencive we give to the model to do what we want

Accuracy: Questions in the training dataset had correct answers, may be math problem

So, we can use rule-base checker to check if the answer is right or wrong.

Formating: Ensuring that the reasoning process is inside the "think" token.

It is also possible to check with rule-base.

③ Group Relative Policy Optimization (GRPO)

↳ A method for updating model parameters based on rewards.

How can we incentivize model to maximize the reward ?

$$\longrightarrow \max_{\theta} \frac{\pi_\theta(o|q)}{\pi_{\theta_{old}}(o|q)} r$$

$\pi_\theta(o|p) \rightarrow$ Current model's probability of generating output (o) in response to question (q)

So, what we do,

- Pick a question & generate an output from the old model.
- Calculate the probability of generating that output for both (old & new) model.

$\boxed{\frac{\pi_\theta(o|q)}{\pi_{\theta_{old}}(o|q)}} \rightarrow$ Deviation of current model from the previous one with respect to that question.

Simply, how much the model changed

Finally "r" is the rule-based reward for the output (o).

As reward is rule-based, no model training or neural network is needed.

Now,

how does maximizing the equation incentivize model to maximize the reward?

If the deviation > 1, the output is more probable to the new model compare to the old one.

1 means a little change may be.

If the reward is > 0, "Good"

< 0, "Bad"

So, the goal is: Update π_θ such that

$$\boxed{\begin{array}{l} \text{deviation} > 1 \text{ for } r > 0 \ \& \\ \text{deviation} < 1 \text{ for } r < 0 \end{array}}$$

However, reinforcement learning can be very unstable. So, the authors used clipping to stabilize training.

$$\max_\theta \ \min\left(\frac{\pi_\theta}{\pi_{\theta_{old}}} r, \ \underline{\text{clip}\left(\frac{\pi_\theta}{\pi_{\theta_{old}}}, 1-\epsilon, 1+\epsilon\right) r}\right)$$

If deviation is too far away from 1, then clip it. "ϵ" is just one parameter that we choose.

Let's say, $\epsilon = 0.3$. If deviation is less than 0.7, clip is going to round it to 0.7 & if it is greater than 1.3, round it to 1.3.

But that is not enough to stabilize & they added regularization.

$$\max_{\theta} \min\left(\frac{\pi_\theta}{\pi_{\theta old}} r, \text{clip}\left(\frac{\pi_\theta}{\pi_{\theta old}}, 1-\epsilon, 1+\epsilon\right) r\right) - \beta D_{KL}\left(\pi_\theta \| \pi_{ref}\right)$$

$$\beta D_{KL}\left(\pi_\theta \| \pi_{ref}\right) = \frac{\pi_{ref}(o|q)}{\pi_\theta(o|q)} - \log \frac{\pi_{ref}(o|q)}{\pi_\theta(o|q)} - 1$$

Actually comparing the probability of output of the current version of the mode to some reference checkpoint of the model that we think good.

Now, aggregating updates over multiple outputs

$$\max_{\theta} \underline{\frac{1}{G}\sum_{i=1}^{G}} \min\left(\frac{\pi_\theta}{\pi_{\theta_{old}}} r, \text{clip}\left(\frac{\pi_\theta}{\pi_{\theta_{old}}}, 1-\epsilon, 1+\epsilon\right) r\right) - \beta D_{KL}(\pi_\theta \| \pi_{ref})$$

So, we are not going to just generate one output for a given question, rather we are going to generate a group of output.

& finally,

rewards are standardized at group level

$$\max_{\theta} \frac{1}{G}\sum_{i=1}^{G} \min\left(\frac{\pi_\theta}{\pi_{\theta_{old}}} A_i, \text{clip}\left(\frac{\pi_\theta}{\pi_{\theta_{old}}}, 1-\epsilon, 1+\epsilon\right) A_i\right) - \beta D_{KL}(\pi_\theta \| \pi_{ref})$$

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots r_G\})}{\text{std}(\{r_1, r_2, \dots r_G\})}$$